Data Dictionary — Nova Scotia Air Quality Analysis

1. Data Source

Source Name: Government of Canada Open Data Portal
Dataset: Air Quality Data (National Air Pollution Surveillance – NAPS)
URL: https://search.open.canada.ca/opendata/?search_text=air+quality

The data originates from the **National Air Pollution Surveillance (NAPS) Program**, Canada's primary source of ambient outdoor air quality data. The program is managed by **Environment and Climate Change Canada (ECCC)** in collaboration with provincial and regional monitoring agencies. NAPS collects continuous air pollutant measurements from monitoring stations across Canada, including Nova Scotia, and publishes quality-assured datasets through the Open Government Portal. The data is used to support air quality trend analysis, environmental reporting, and public health indicators such as the Air Quality Health Index (AQHI). [open.canada.ca], [canada.ca]

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2. Model Overview

The dataset was modeled in **Power BI** using a **star-schema style design** to support analytical reporting and time-based trend analysis.

- The central **fact table** stores daily air quality observations.
- Supporting **dimension tables** provide descriptive context for time, monitoring stations, and pollutants.
- A dedicated **Date dimension** was created and marked as the official date table to enable time-intelligence calculations (YTD, MoM, YoY).
- Relationships are one-to-many from dimension tables to the fact table.
- Aggregated insights are produced using DAX measures rather than calculated columns.

This model supports slicing and filtering air quality trends by **date, station, and pollutant** while maintaining reporting performance and clarity.

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3. Fact Table

Fact_AirQuality

Column Name	Description

Date	Calendar date of the air quality observation (daily level).
Station	Name of the air monitoring station in Nova Scotia.
Pollutant	Pollutant code measured (e.g., O3, NO2, SO2).
Concentration	Daily average pollutant concentration measured at the station (ppb).

Grain:
One record per **day**, per **station**, per **pollutant**.

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4. Dimension Tables

DimDate

Column Name	Description
Date	Unique calendar date (primary date key).
Year	Calendar year.
Quarter	Calendar quarter (Q1–Q4).
Month	Numeric month (1–12).
Month Name	Descriptive month name (January–December).
Season	Seasonal classification derived from month.

Notes:

- Marked as the official Date table in Power BI.
- Used for time-intelligence DAX calculations and hierarchical drill-downs.

DimStation

Column Name	Description
Station	Name of the air quality monitoring station.

Notes:

- Provides geographic and monitoring context.

- Used for station-level comparison and ranking visuals.

DimPollutant

Column Name	Description
Pollutant	Short pollutant code (O3, NO2, SO2).
Pollutant Name	Full pollutant description (e.g., Ozone, Nitrogen Dioxide).
Unit	Measurement unit for concentration values (parts per billion – ppb).

Notes:

- Enriched manually to improve interpretability and reporting clarity.
- Enables pollutant-specific trend and comparison analysis.

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5. Analytical Usage

The model supports:

- Trend analysis over time (monthly, quarterly, yearly)
- Comparison of pollution levels across stations
- Pollutant-specific analysis and composition
- Time-intelligence measures such as YTD averages and year-over-year change
- KPI reporting using DAX measures